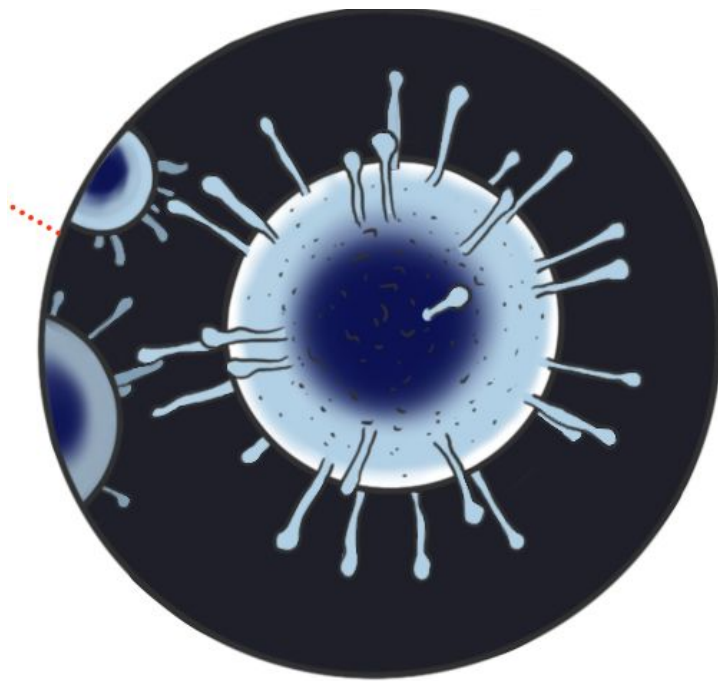


PLABABLE

GEMS 

VERSION 1.3

INFECTIOUS DISEASES



Respiratory Infections

Community acquired pneumonia (mild)	Amoxicillin
Community acquired pneumonia (moderate)	Amoxicillin + Clarithromycin
Community acquired pneumonia (severe)	Co-amoxiclav + Clarithromycin Co-amoxiclav (amoxicillin + clavulanic acid)
<i>Pneumocystis Jirovecii</i> pneumonia (prev as “ <i>P. Carinii</i> ”)	Co-Trimoxazole (trimethoprim + sulfamethoxazole)
Tuberculosis (TB)	First 2 months: RIPE ● Rifampicin ● Isoniazid ● Pyrazinamide ● Ethambutol Next 4 months: RI Rifampicin and isoniazid
Aspiration pneumonia (community acquired)	Amoxicillin + Metronidazole

Central Nervous System Infections

Out of hospital meningitis	Benzylpenicillin
In-hospital meningitis	Cefotaxime IV or Ceftriaxone IV
Listeria meningitis	Amoxicillin or Ampicillin + Gentamicin
Cryptococcal meningitis	Amphotericin B
Meningitis prophylaxis (for contacts)	1st line: Ciprofloxacin 2nd line: Rifampicin

Meningitis

Brain trainer:

A woman over the age of 60 has meningitis in the Emergency Department. What should be administered empirically?

→ IV ceftriaxone and IV amoxicillin

If she was below 60, it would have just been IV ceftriaxone

Meningitis

Brain trainer:

A person with headache, neck stiffness, photophobia and high fever. What is the most likely organism?

If gram positive diplococci seen

➔ ***Streptococcus pneumoniae***

If gram negative diplococci seen

➔ ***Neisseria meningitidis***

If non-blanching rash seen

➔ ***Neisseria meningitidis***

Genitourinary Infections

Infection	Antibiotic
Lower uncomplicated UTI	Trimethoprim or Nitrofurantoin
<i>Candida albicans</i> (Vulvovaginal Candidiasis)	Clotrimazole or Fluconazole
<i>Trichomonas vaginalis</i> (Bacterial vaginalis)	Metronidazole
Cervicitis (Chlamydia)	1st line: Doxycycline 2nd line/pregnant: Azithromycin
Cervicitis (Gonorrhoea)	Ceftriaxone IM or Ciprofloxacin (if sensitivity is known)
Pelvic inflammatory diseases	1st line: Ceftriaxone + Metronidazole + Doxycycline 2nd line: Metronidazole + (Ofloxacin or Moxifloxacin)
Syphilis	Penicillin G
Genital herpes (HSV)	Aciclovir

Weakened Immunity

Brain trainer:

A pregnant woman has thick white marks inside her mouth. She is a smoker. What is the most likely diagnosis?

➔ **Candidiasis**

A weakened immunity (due to pregnancy) and a more common diagnosis speaks for candidiasis.

If the stem contained “cannot be rubbed off”, choose leukoplakia.

Gastrointestinal Infection

Infection	Antibiotics
Salmonella / Shigella / Campylobacter	Erythromycin OR Azithromycin OR Clarithromycin OR Ciprofloxacin
<i>Clostridium difficile</i> (Pseudomembranous colitis)	For mild: Oral Metronidazole For severe: Vancomycin
<i>Helicobacter pylori</i>	OAC or OAM Omeprazole Amoxicillin Clarithromycin or Metronidazole

Ear, Nose & Throat infection

Acute bacterial otitis media

Amoxicillin

Upper respiratory tract infection:

- Pharyngitis
- Tonsillitis
- Laryngitis

Phenoxymethylpenicillin

Laryngitis

Brain trainer:

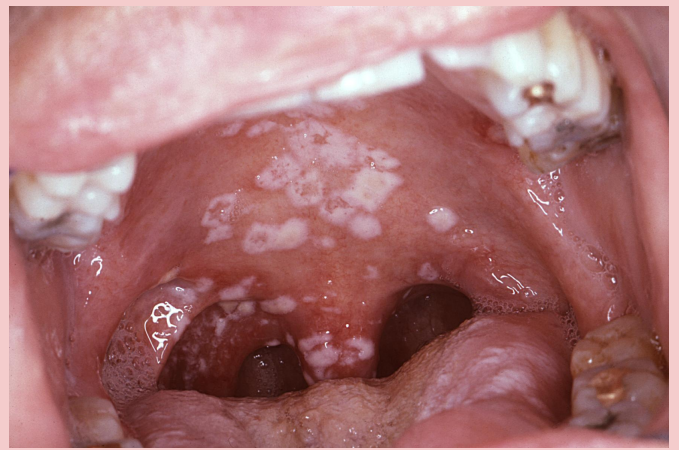
A boy has a 2 day history of hoarseness of voice, dry cough, fever and malaise. On examination his vocal cords are oedematous. What is the most appropriate investigation?

→ None required

Skin, Joint and Other Infections	
Cellulitis	Flucloxacillin
Mastitis	
Diabetic Foot infection	
Scabies	Permethrin 5% topical
Septic Arthritis	Flucloxacillin ± IV Gentamicin or Metronidazole
Osteomyelitis	
Toxoplasmosis	Pyrimethamine + Sulfadiazine

Oral Thrush (Oral candidiasis)

Features:



- Thick white marks
- $\pm$ inflamed mouth / tongue
- Can be **rubbed** out
- $\pm$ Red inflamed painful sore mouth angles

Contributing factors

- History of immunosuppression -diabetes, recent use of antibiotics & regular steroids eg. asthmatic
- Smoking
- Elderly
- Pregnancy

Treatment

- Stop smoking
- Good inhaler technique and hygiene eg. use a spacer and rinse mouth after use
- 1st line: oral miconazole gel
- Oral fluconazole 50mg OD 7 days in severe cases

Leukoplakia

Features

- Raised edges
- Bright white patches
- Sharply well defined
- Cannot be rubbed out



Contributing factors

- History of smoking

Investigation

- Biopsy (they are premalignant)

Treatment

- Smoking cessation
- Observe or surgical excision depending on biopsy result

Streptococcus Pneumoniae

Facts:

- The **most common cause** of pneumonia
- A gram +ve diplococci

Features:

- Productive cough
- Fever 🤒
- Chest tightness
- Unilateral basal crackles on auscultation
- Unilateral lobar consolidation on x-ray

It is associated with *Herpes Labialis*

Different Types Of Pneumonia

Pneumococcal streptococcal Also associated with HIV patient - CD4>200 !!	● Herpes Labialis ● Typical features of community acquired pneumonia → Productive cough → Fever → Unilateral basal crackles → Unilateral basal consolidation ● Lobar consolidation
Mycoplasma	● Erythema multiforme ● Atypical features: → Young adult → Dry cough → Bilateral consolidation ● Patchy consolidation of 1 lower lobe
Pneumocystis jirovecii	● HIV with CD4 < 200 ● <u>±</u> desaturation on exercise ● Dry cough ● Bilateral consolidation ● Exertional dyspnoea

Type Of Pneumoniae

Brain trainer:

A HIV man presents with symptoms of pneumoniae. His CD4 count is measured at 350mm^3 . X-ray shows shows lobar consolidation. What is the most likely causative organism?

→ **Streptococcus pneumoniae**

Type Of Pneumoniae

Brain trainer:

A 14 year old child presents with meningitis. CSF sample collected appears purulent. What is the most likely causative organism?

→ **Streptococcus pneumoniae**

Different Types Of Pneumonia

<i>Staph. aureus</i>	● Pneumonia developed after influenza (flu) ● Common in IV drug abuser and elderly ● Bilateral cavitation
<i>Legionella</i>	● Pneumonia developed after exposure to water, staying in hotel ● Low sodium ● Low lymphocytes ● Bibasal consolidation ● Macrolides/Tetracyclines are usual treatment of choice
<i>Klebsiella</i>	● Cavitating pneumonia ● Particularly upper lobe

Type Of Pneumoniae

Brain trainer:

An IV drug users presents with symptoms of pneumoniae. One week earlier he had flu like symptoms. X-ray shows bilateral cavitations. What is the most likely causative organism?

→ **Staphylococcus aureus**

Tuberculosis

Facts:

- Caused by *mycobacterium tuberculosis* (Acid fast bacilli)
- Travelling Hx: **South Asia, Sub-saharan Africa, and India**

Diagnosis:

1. **Sputum staining** for acid-fast bacilli
2. **Bronchoalveolar lavage** if no sputum
3. **Gastric lavage** if above not available (sputum might be swallow by patient during sleep)

High risk factor groups:

- Homeless
- Drug abuser
- Smoker
- Low socioeconomic class

Tuberculosis

Features:

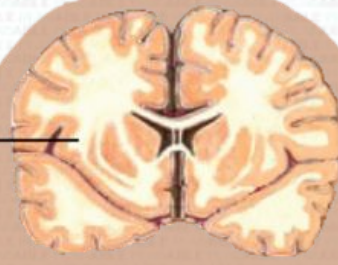
- **Chronic productive cough**
- **Hemoptysis**
- **Weight loss (Cachexia)**
- **Fatigue**
- **Night sweats**
- **Small area of caseating granulomas**
- **Upper lobe consolidation, infiltration with cavitation on chest x-ray**
- **Initially **tender and firm** then discrete on palpation**

Tuberculosis

Main symptoms of Pulmonary tuberculosis

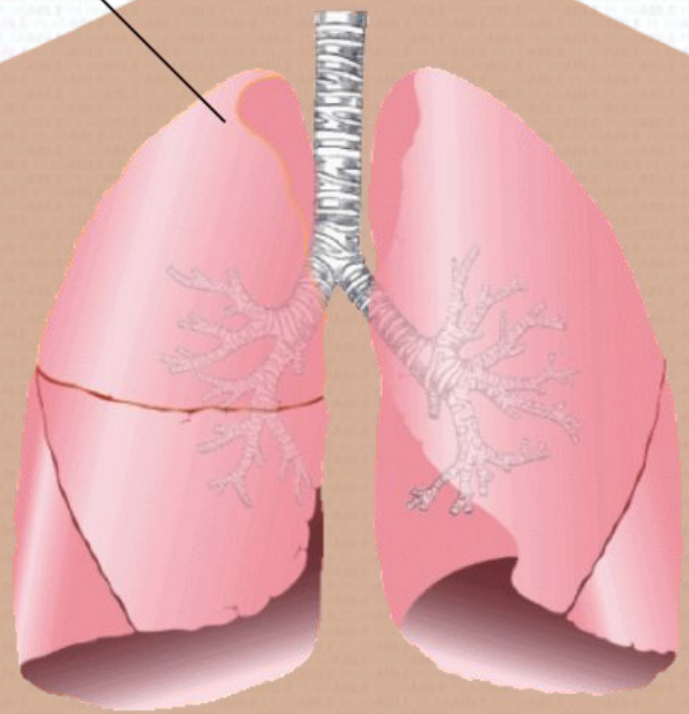
Central

- appetite loss
- fatigue



Lungs

- chest pain
- coughing up blood
- productive, prolonged cough



Skin

- night sweats,
- pallor

Tuberculosis

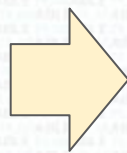
Screening for contacts:

- For **latent TB not acute TB**
- **Mantoux test** for contact who **have not been vaccinated with BCG** before
- **Interferon gamma test** for contacts who **have been vaccinated with BCG** before

Treatment

1st 2 months - **RIPE**

Rifampicin,
Isoniazid,
Pyrazinamide,
Ethambutol



Next 4 months - **RI**
Rifampicin, **I**soniazid

These are not contraindicated during pregnancy
but streptomycin is!!

Tuberculosis

Directly-Observed Therapy:

For patient who need support to manage TB in outpatient setting:

For underserved groups

- Homeless
- Imprisoned
- Drug or alcohol misuse
- Patient who have not been adherent to therapy
- Patient who are too ill to adhere to therapy

Toxoplasmosis can present with **splenomegaly** and **cervical lymphadenopathy**
Weight loss is usually NOT SEEN.

Tuberculous Lymphadenitis

Contributing factors:

Travelling history - particularly to/from India

Features:

- Fever
- Cough
- Cervical lymphadenopathy
- Caseating granuloma in LNs

Sarcoidosis and Crohn's disease have
NON-caseating granuloma

Laryngeal TB

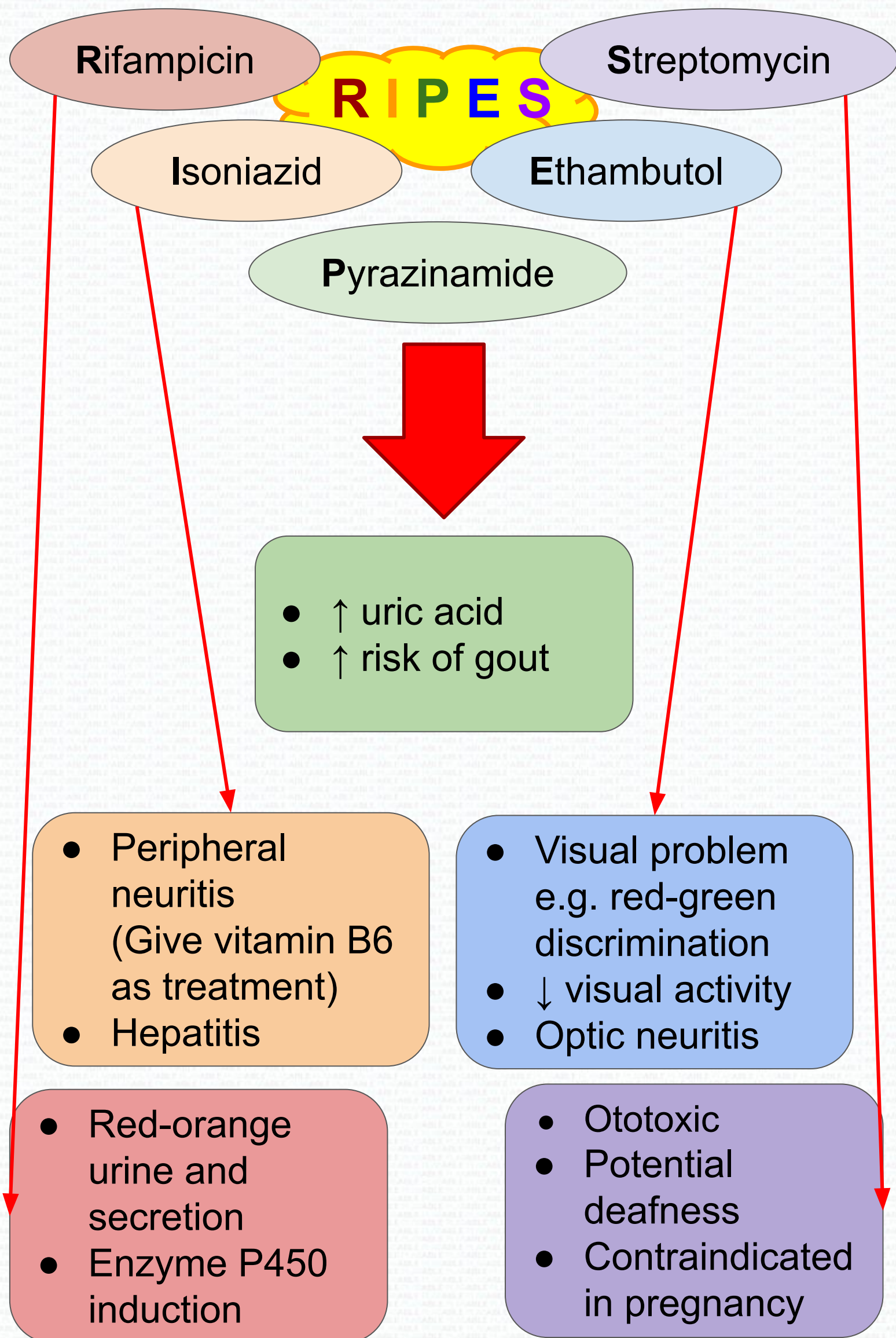
Contributing factors

- IV drug abuser
- Low socioeconomic

Features

- Fever
- Cough
- Cervical lymphadenopathy
- Hoarseness
- Dysphagia
- Weight loss

Side Effects of Anti-TB drugs



Tuberculosis

Brain trainer:

Which medication for the treatment of tuberculosis is contraindicated in pregnancy?

→ Streptomycin

Tuberculosis

Brain trainer:

An homeless man has tuberculosis. What is the most appropriate management of this patient?

→ **Directly observed therapy**

Scabies

Facts:

- Organism is *Sarcoptes scabiei*
→ A parasite causes skin infestation
- Transmitted by **skin to skin contact**
- Causes **allergic reaction**
→ Pruritus **NOT** infection

Features:

- Linear tracks on skin (**Burrows**)
- **Severe pruritus** (particularly night time)
- Particularly in area of skin folds, flexures of:
 - Wrists
 - Finger webs
 - Elbow
 - Axilla
 - Areola
 - Genitalia



Treatment:

- Permethrin 5% cream
- Simultaneously treat all close contacts

Hint: Nursing home!!

Scabies

Brain trainer:

What is the mechanism of itching in a scabies infection?

→ Allergic reaction

Infectious Mononucleosis (IMN)

Glandular fever

Cause:

- **Epstein-Barr virus (EBV)**
aka (Human Herpesvirus 4 HHV-4)

Features:

- Sore throat
- Exudative tonsillar enlargement
- Fever
- Malaise
- Lymphadenopathy → particularly cervical
- ± Splenomegaly
- ± Palatal petechiae

Diagnosis:

- Heterophil antibody test
= Monospot test = Paul Bunnell
- FBC: ↑ WBC and ESR
- Lymphocytosis → Atypical lymphocytes > 20%

Treatment:

- Supportive i.e. simple analgesic for pain

Giving **Amoxicillin** and **Ampicillin** can lead to development of **pruritic maculopapular rash**

Gastroenteritis

Features:

- Nausea
- Vomiting
- Diarrhea ± blood
- Abdominal pain

Investigations:

- Stool microscopy, culture and sensitivity
- FBC, renal function and electrolytes

Management:

- Good hygiene ± isolation
- Hydration + small, light meals
- When indicated → antibiotics

Important note:

Most cases can be managed conservatively (i.e. without investigations or antibiotics). Exceptions include bloody diarrhea, immunocompromised + prolonged (7 days+) cases.

In UK, patients are allowed to return to work **48 hours** after the **last episode of symptoms (Diarrhoea or vomiting)**

Gastroenteritis

Features:

- Diarrhoea
- Vomiting

In UK, patients are allowed to return to work **48 hours** after the **last episode of symptoms (Diarrhoea or vomiting)**

Investigations:

- Heterophile antibodies:
 - Paul Bunnell
 - Monospot
- EBV specific antibodies

Treatment:

- Supportive measures e.g. paracetamol and ibuprofen for pain / fever

Meningitis

Otitis media

Meningitis

Hearing loss



Delayed complication

After treating,
MUST arrange a hearing test

Kaposi Sarcoma

It is a cancer of connective tissue most commonly seen in immunocompromised patients (HIV / AIDS)



Blood vessels ↑ in size



Features:

- Red, purple, brown or black nodules or papules
- Usually non-painful
- Mouth, nose or throat are common sites
- Can grow internally (eg. lungs, GI tract)

Kaposi Sarcoma

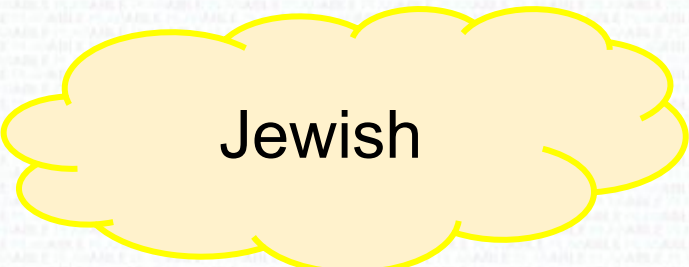
Risk factors



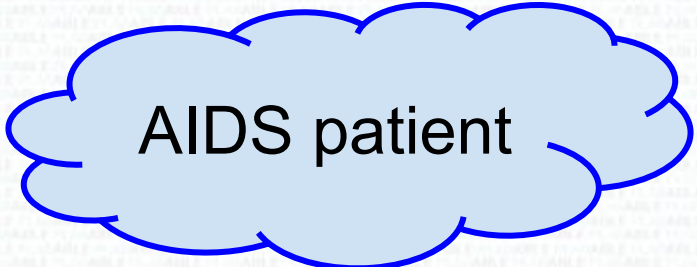
Homosexual



Mediterranean



Jewish



AIDS patient



Bisexual

Chicken Pox

Caused by varicella zoster virus

Infectious route:

- Very contagious
- Mainly via **respiratory route - airborne**
- Can be transmitted by direct contact with vesicles
- **Dried out and crusted** vesicles **CANNOT** transmit



Infectious period:

- 4 days **BEFORE** rash
- 5 days **AFTER** rash first appeared
- Stop when vesicles dried out and crusted

Features:

- Fever (38-39 °C)
- Pruritic rash (itchy)
- Rash:
 1. Macules
 2. Papules
 3. Vesicles
 4. Dry crust
- Usually starts on face or trunk and then may spread over entire body



Chicken Pox - Management

< 12 years old

Reassurance + supportive measures.

- Paracetamol for fever
- Antihistamines and calamine lotion for itching

Oral antibiotic if superimposed infection is suspected:

- Fever
- Discharging pustules
- Redness around vesicle
- Pinkish fluid secretion
- High fever

When can children go back to school after chicken pox?

After vesicles dried out and crusted
Usually 5 days after onset

Chicken Pox - Management

Varicella-zoster
Immunoglobulin
(VZIG)

For Pt with exposure:

- Immunocompromised
- Pregnant with no VZ antibodies
- Newborn with peripartum exposure

e.g. On chemotherapy, on long
term steroid
= Immunocompromised

Aciclovir

For Pt who develops
chicken pox:

- Immunocompromised
- Pregnancy

Chicken Pox

Brain trainer:

An adult has chicken pox with a fever of 39.1 and pinkish white thick fluid is seen secreted from a few of the lesions. What is the most appropriate medication to prescribe?

→ Oral antibiotics

Chicken Pox

Brain trainer:

An immunocompromised man reports that his partner currently has chicken pox. His history is positive for chicken pox in childhood. What is the most appropriate action?

→ Obtain serology for varicella immunity

Shingles

Facts:

- It is caused by varicella zoster virus (VZV)
- It is a reactivation of VZV especially in immunocompromised / old patients
- Presented as chicken pox at initial VZV infection
- Virus remain inactive in nerve cells

Features:

- Painful skin rash with blisters in localized area



Management:

- Obtain serology for varicella immunity in immunocompromised patients
- Immunocompromised- HIV positive, diabetic, on long term steroids

Treatment:

- Aciclovir
- Pain management

Ramsay Hunt Syndrome (Herpes zoster oticus)

Facts:

- It is a reactivation of VZV
- Localised in geniculate ganglion of the facial nerve (7th CN)

Features:

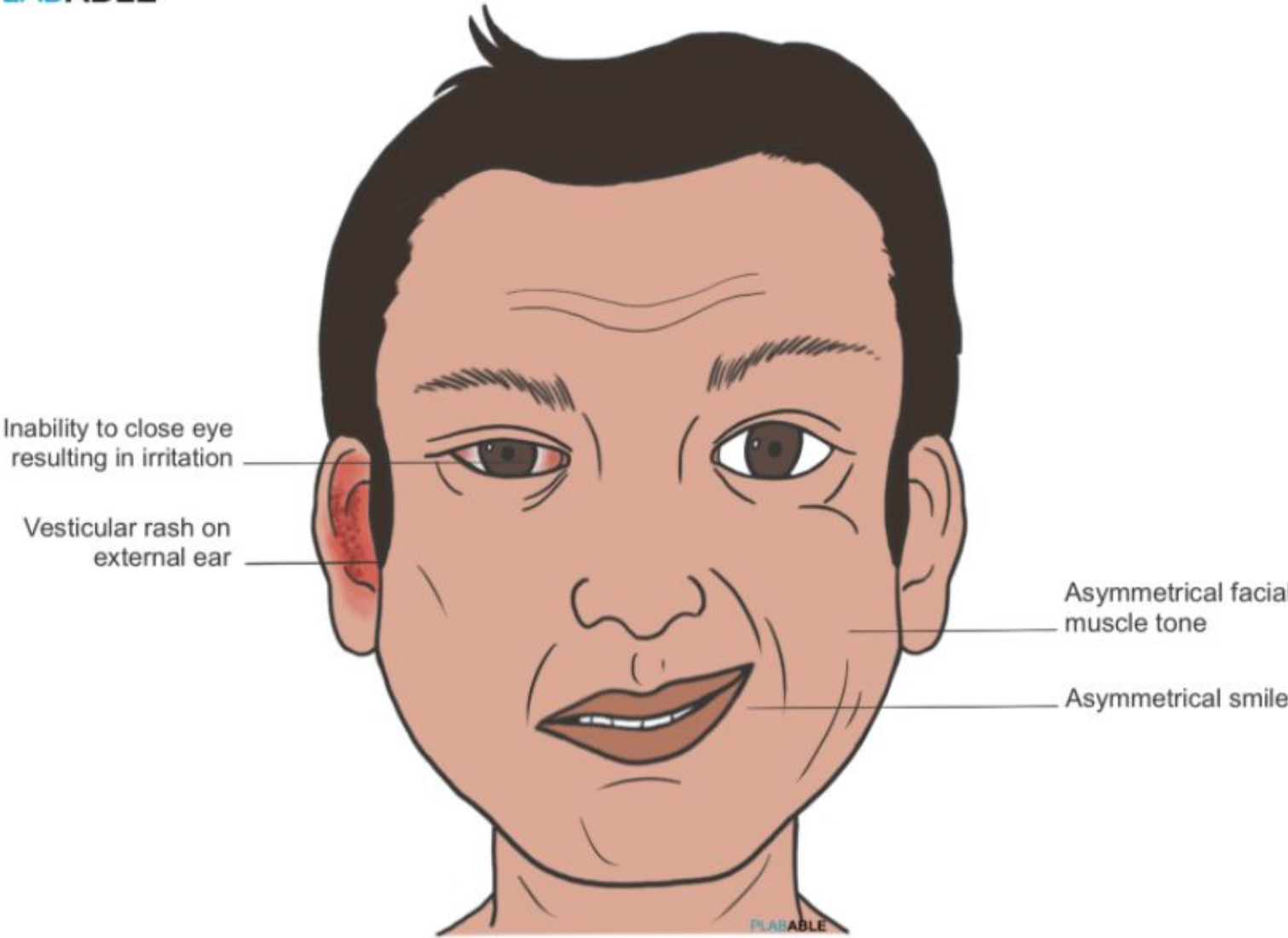
- Facial palsy (ipsilateral facial palsy, loss of taste)
- Otalgia - ear pain
- Tinnitus
- Vertigo
- Unilateral hearing loss
- Painful rash or vesicles around ears or on auditory canal

Treatment:

- Aciclovir
- Corticosteroid
- Amitriptyline for pain

Ramsay Hunt Syndrome (Herpes zoster oticus)

PLABABLE



Ramsay Hunt Syndrome

Herpes Zoster Ophthalmicus

Facts:

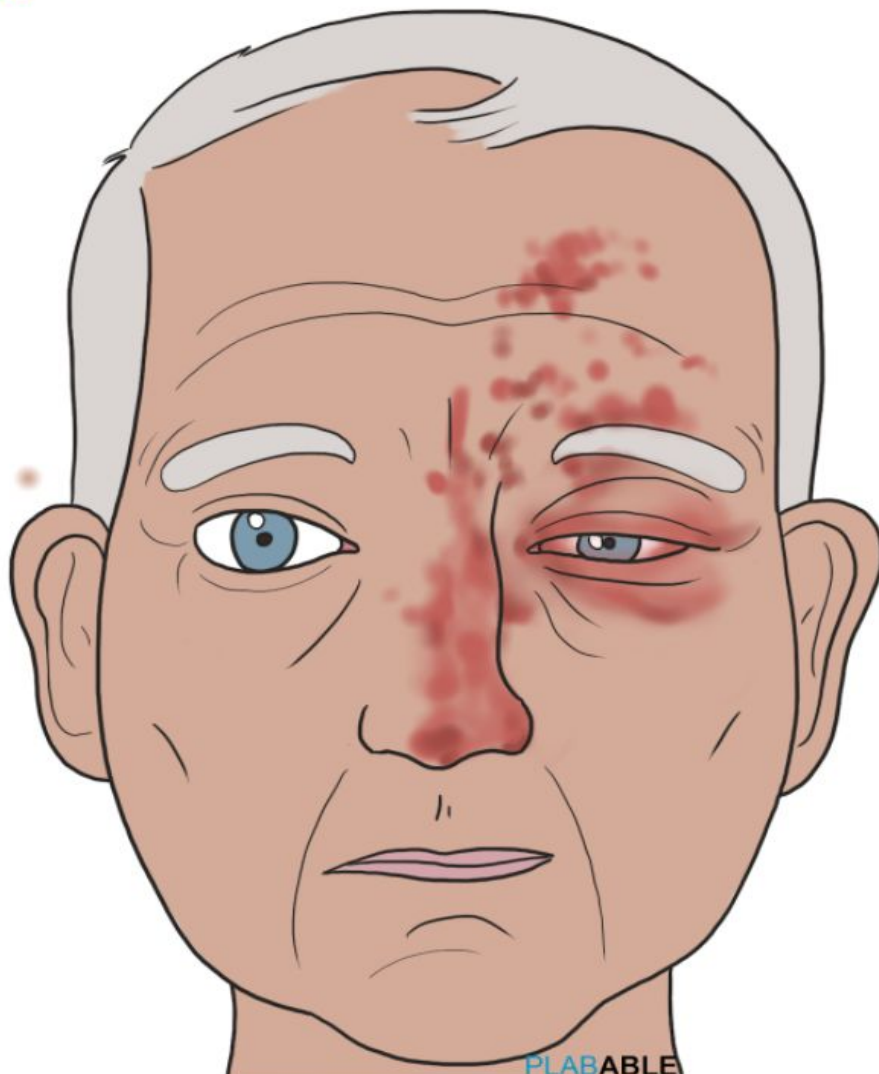
- It is a reactivation of VZV
- Localised in ophthalmic branch of trigeminal nerve (5th CN)

Features:

- Conjunctivitis
- Keratitis
- Painful rash or vesicles around eyes

Treatment: Aciclovir

PLABABLE



Hutchinson's Triad

PLABABLE

Herpes Zoster Ophthalmicus

Brain trainer:

A patient has suspected shingles with a rash present on the tip of the nose. What is this finding called? and what is its significance?

→ Hutchinson's sign → ocular involvement

Lyme disease (Lyme borreliosis)

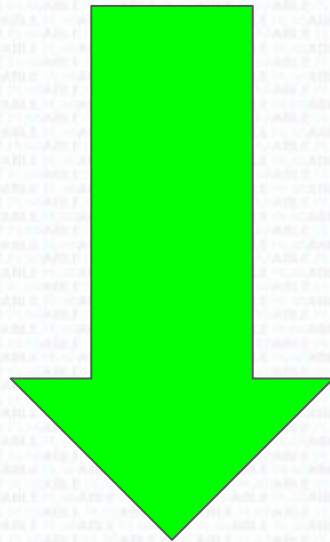
History of camping or walking in jungles or gardens >>> **Ticks bite**

Beginning stage:

Erythema migrans (erythematous, painless, non-itchy)

± fever, headache, myalgia and general aches)

It could present as an **annular rash** with scaly edge, slow growing with general aches and pain



Later stage:

- Facial paralysis
- Meningitis
- AV heart block
- Myocarditis
- Arthritis

Lyme Disease (Lyme borreliosis)

Diagnosis:

- Check for antibodies to **Borrelia- Burgdorferi**

Treatment:

- **Doxycycline** or **amoxicillin** if contraindicated
e.g. pregnancy
- **Ceftriaxone** in disseminated disease

Lyme Disease

Brain trainer:

A pregnant woman has lyme disease in the early stages. What is the most appropriate management?

→ Amoxicillin

Meningitis

In Babies And Toddlers

Fever,
cold
hands
and feet

Refusing
food and
vomiting

Fretful,
dislike
being
handled

Rapid
breathing
and
grunting

Unusual
crying,
moaning

Stiff neck,
dislike
bright
lights

Pale,
blotchy
skin,
spot/rash

Tense,
bulging
fontanelle
(soft spot)

Drowsy,
floppy,
unresponsive

Convulsion
or seizures



Meningitis in babies and toddlers

Fever,
cold
hands
and feet

Vomiting

Pale,
blotchy
skin,
spot/rash

Confusion
and
irritability

Drowsy,
difficult to
wake



Severe
headache

Stiff neck

Severe
muscle
ache

Dislike
bright lights

Convulsion
or seizures

Meningitis

Vomiting

Fever

Hx of seizures

Altered mentation

Photophobia

Headache

Diagnostic investigation

Without rash

With rash

Lumbar puncture
= CSF analysis

Blood culture

Contraindicated to:

- ↑Intracranial Pressure
- Bulging, tense fontanelle
- Ongoing seizures
- GCS > 9 or a drop of ≥ 3
- Unequal, dilated, unresponsive pupils
- Papilledema

Check for
Meningococcal septicemia

→ *Neisseria meningitidis*

Meningitis

Treatment for meningitis



ASAP

Health protection team must be notified immediately once there is a clinical suspicion of meningitis!!!

In exam, look out for the points below:

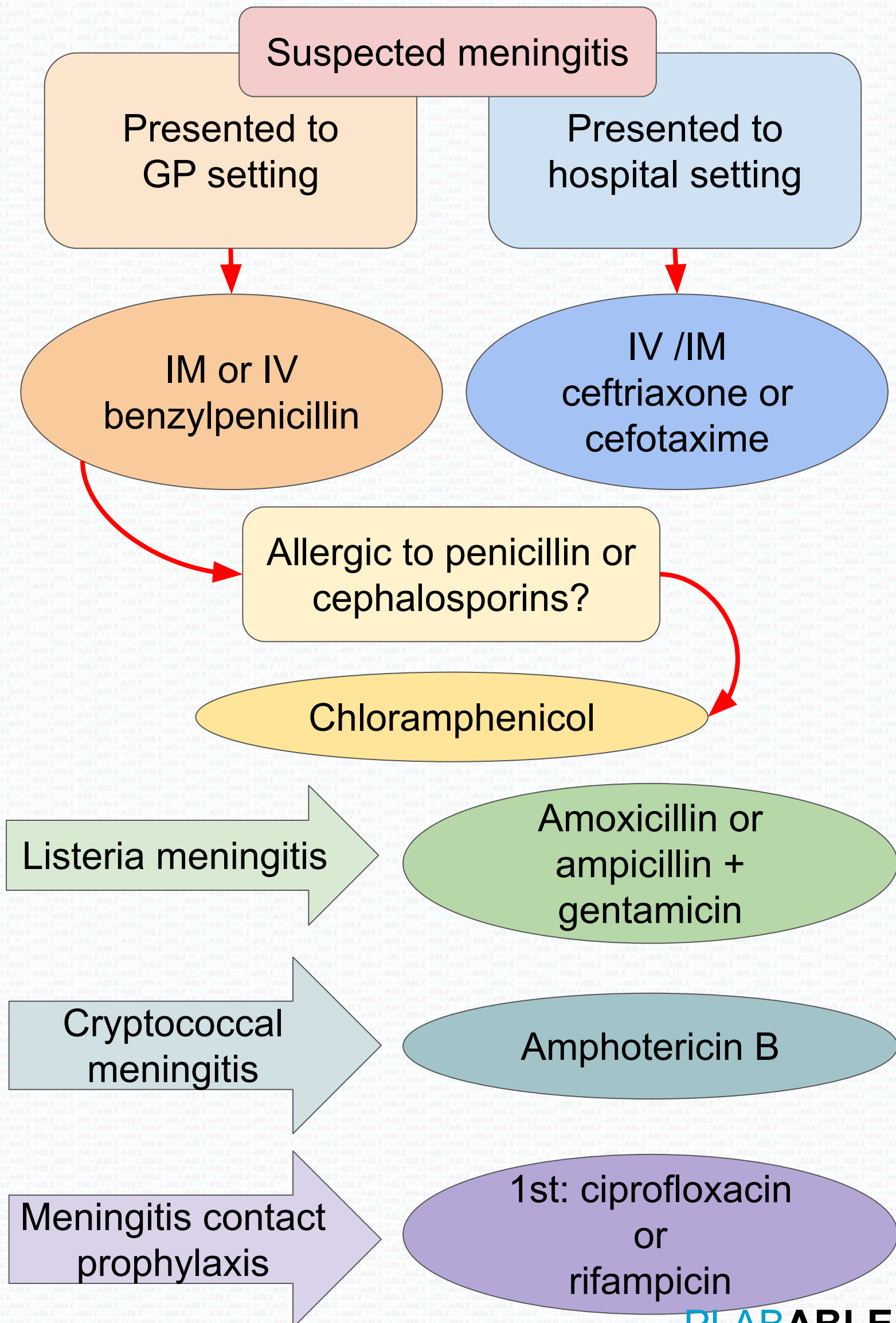
- Arthralgia and muscle aches
- Cold periphery
- Pale or mottled skin
- SOB
- Rash

- Photophobia
- Severe headache
- Neck stiffness

Septicemia

Meningitis

Meningitis Treatment



Schistosoma

Features:

- Hx of travelling to **Africa** or SE Asia
- Pruritic skin rash at site of penetration
- Fever, hives, headache, weight loss
- $\pm$ Hepatomegaly
- Hematuria (dark and red urine)

Schistosoma organisms

*Schistosoma
Mansoni*

*Schistosoma
Haematobium*

Affects intestine
and liver

Affects urinary
bladder

Hepatomegaly

Hematuria, UB
calcification and
obstructive
uropathy

These features
are caused by 2
different species

Schistosoma Haematobium

Features:

- Hx of travel to Africa
- Haematuria
- Urinary bladder calcification
- Ulceration and obstructive uropathy
- Increased bladder cancer risk

Investigations:

- X-ray:
 - Urinary bladder calcification and obstructive uropathy
- Ultrasounds:
 - Hydronephrosis
 - Thickened bladder wall

Important notes:

- Can lead to bladder cancer
- May be after up to 20 years post-infection

Similar organisms:

- *Schistosoma mansoni*
 - Affects intestines and hepatosplenic
 - Hepatomegaly

Malaria

Features:

- **Hx of travel to Africa**
- Intermittent fevers
- Rigours
- Headache
- Malaise
- Cough
- Myalgia
- Gastrointestinal upset
- Hepatosplenomegaly
- Jaundice

Investigations:

- Microscopy of thick and thin blood film (most accurate)
- Full blood count may show anaemia and thrombocytopenia
- Rapid diagnostic test → Used as adjunct to blood films and not a replacement

Important note:

Avoid the belief that prophylaxis prevents all malaria

Cerebral Malaria

Features:

- Hx of travelling to **Africa**
- Meningitis-like symptoms:
 - Fever and chills
 - Neck stiffness
 - Vomiting
 - Impaired consciousness
- **Anaemia**
 - Differentiate from meningitis



To suspect!

- Hx of travel to malaria affected area(s)
- In the past year particularly over the **last 3 months**

Malaria prophylaxis does not guarantee full protection against all subtypes of malaria

Treatment For Malaria

1. Primaquine

- If blood film shows ring form *Plasmodium* with schuffner's dots
 - **To be used at latent/dormant stage of Plasmodium ovale or vivax latent hypnozoites in liver**
 - Targets and eradicate all stages - liver latent and RBCs
 - Contraindicated to patients with G6PD as it can cause severe hemolysis
 - Contraindicated to pregnancy and breastfeeding
- *MUST** screen for G6PD deficiency before commencing on therapy*

2. Chloroquine

- In non-falciparum / non-hypnozoite malaria
- **To be used in active stage** and targets RBCs
- Can be used in pregnancy
- Not effective in chloroquine-resistant area(s) e.g. Sub-Saharan Africa

3. Quinine

- To be used when chloroquine failed

4. Mefloquine

- To be used if travelling to chloroquine-resistant area(s)
- Can be used in pregnancy

Treatment For Malaria

Brain trainer:

An african woman has malaria. Blood film shows ring form plasmodium with schuffner's dots in red blood cells. What is the SINGLE most appropriate drug to eradicate this infection?

→ Primaquine

Needle Stick Injuries

Being pricked by a needle

Basic 1st aid

- Washing with soap under running water
- Encouraging bleeding in affected area

Request patient's permission to investigate any blood-borne infection
(HIV, HCV and HBV)

Low risk:

- Safe sexual intercourse
- Does not use IV drugs

Test healthcare professional for **hepatitis B surface antigens**

High risk:

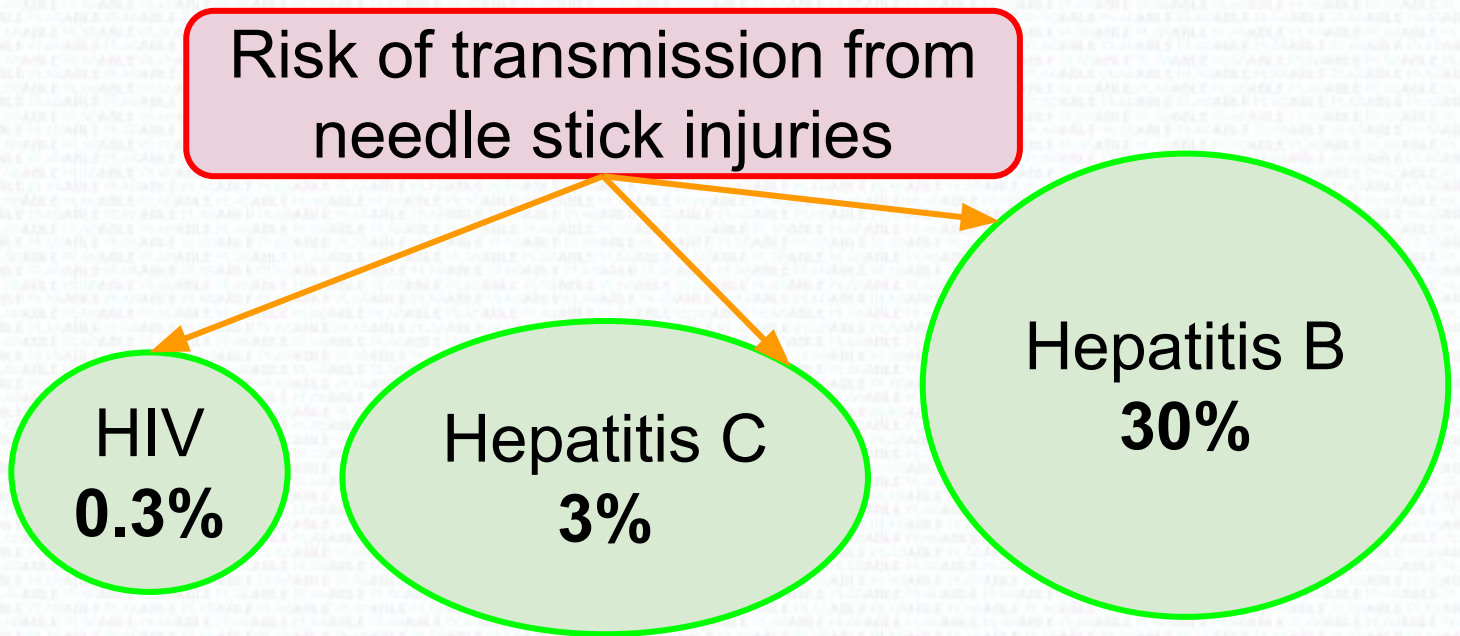
e.g. IV drug abuser, drug addicts

Start **post-exposure prophylaxis (PEP)**

Offer hepatitis B booster

- If booster are no received previously
- If healthcare professional cannot remember when last dose was

Needle Stick Injuries



Healthcare professionals should return in **6 weeks** and **check for HIV and HCV**

Patient should always be **tested for HIV, HCV and HBP**

Healthcare professional should always be tested for **hepatitis B surface antigen** and **offer hepatitis B booster** if cannot remember when was last dose in **low risk injuries**

PEP should be offered to healthcare professionals if injuries is **high risk**

Surgical Prick Injury

Brain trainer:

A surgeon is pricked by a needle used during an appendectomy. What action should be taken?

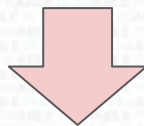
→ Patient: test HBV, HCV, HIV

→ Surgeon: store blood, test HBV, offer HBV booster

HIV Post-Exposure Prophylaxis (PEP)

Antiretroviral medications to be given as soon as possible after exposure:

- **Non-safe sexual intercourse** with high risk individual
- **Needle stick injury** when from:
 - High risk source
 - High risk bite e.g drug addict



To be start ASAP
1-2 hours, up to **72 hours** after exposure



To complete a 28 days course of PEP

If human bite: co-amoxiclav 7 days or metronidazole + doxycycline if penicillin allergic



Recheck HIV viral load after 6 months
Aim: viral load < 200

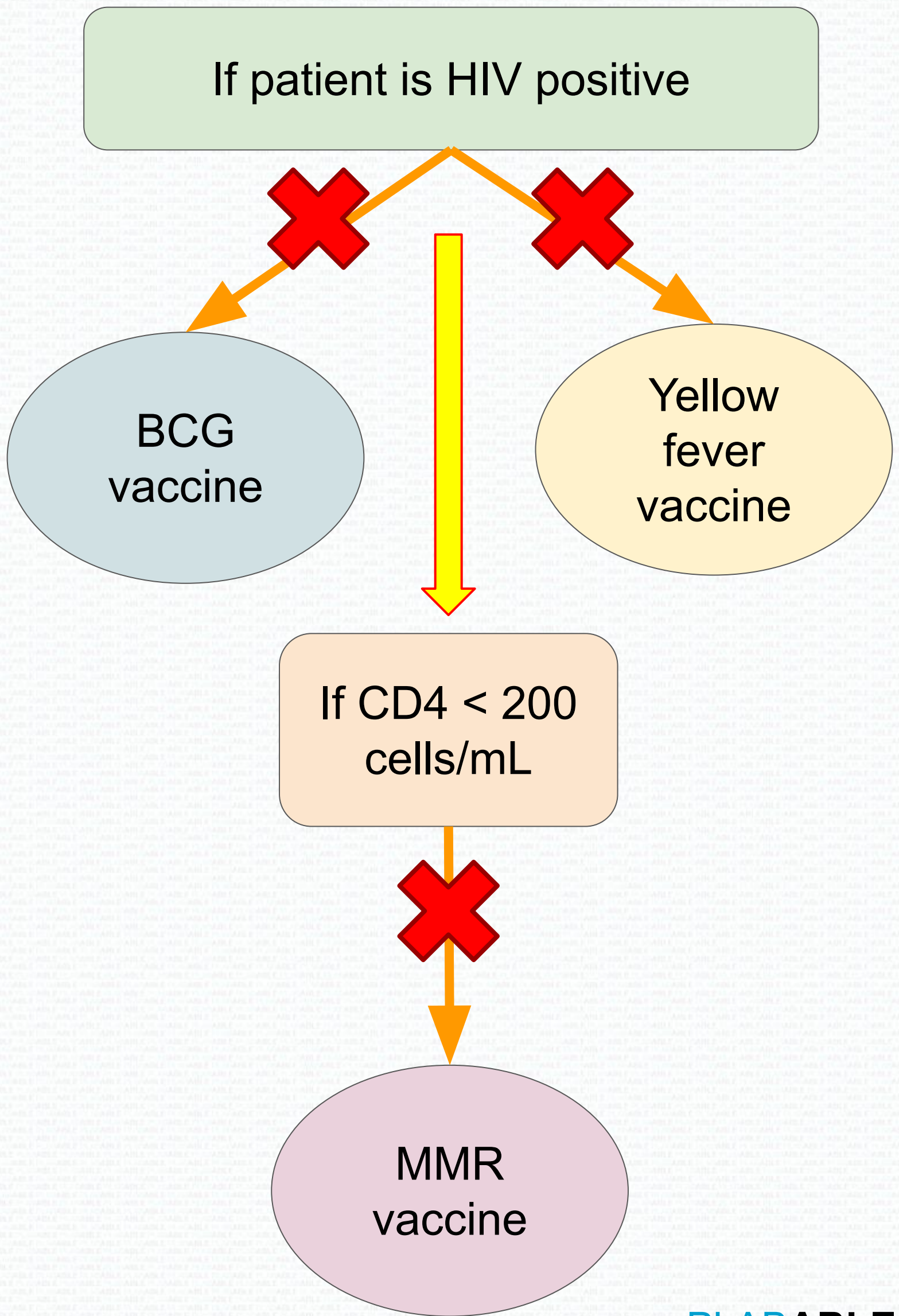
Bite Injury

Brain trainer:

A police officer is bitten by a drug user. What would you offer the police officer?

→ Post-exposure prophylaxis

Vaccine To Avoid In HIV Patients



Vaccine To Avoid In HIV Patients

Brain trainer:

How is the vaccination schedule for a HIV positive infant?

→ All vaccines as scheduled except BCG vaccine

Tetanus Prophylaxis (after injury)

1. Is wound high risk - dirty, contaminated or compound fracture?

Yes

- Give IM human **tetanus immunoglobulins**
- Regardless of the immunization status

No

- No need for IM human tetanus immunoglobulins



2. Person's immunization status?

Fully immunised and up-to-date
i.e. completed 5 doses of tetanus vaccine

No need for tetanus vaccine

Unknown or incomplete

- Give complete course of tetanus vaccine (5 doses)
- Or full course of diphtheria, tetanus and pertussis vaccine

Sometimes we give Abx for wound with high risk of infection

How To Give Tetanus Vaccine

Adult who has not been immunised

- 5 doses required
- First 3 doses should be given 1 month apart
- Remaining 2 are boosters
- 1st booster give at **10** years after primary course
- 2nd booster give at **10** years after first booster

Children (below 10) who has not been immunised

- 5 doses required
- First 3 doses should be given 1 month apart
- Remaining 2 are boosters
- 1st booster give at **3** years after primary course
- 2nd booster give at **10** years after first booster
- **Can be given as diphtheria, tetanus, pertussis combined vaccine**

Remember:

1. Dirty wound? → Immunoglobulines
2. Immunisation status? → vaccine
3. High risk infection? → antibiotics

Tetanus Prophylaxis

Brain trainer:

A man presents with a deep penetrating wound after stepping on a nail which was heavily contaminated with soil. He has completed a full priming course of tetanus vaccine with the last dose within 10 years. What is the most appropriate management to be given?

→ Reassure

Mumps - Paramyxovirus

Facts:

- Contagious and infectious virus
- Transmitted via close contact e.g. droplet of saliva

Features:

- Bilateral parotitis: pain and tender swelling at angles of jaw - periauricular
- Swelling of parotid salivary gland
- Fever
- Dry mouth - due to blockage of salivary gland
- Difficulty in opening mouth or talk - due to swelling
- Orchitis (4-5 days post parotitis) - **NOT always**
 - Severe testicular pain
 - Swollen edematous scrotum
 - Impalpable testes
 - Risk of sterility in males

Treatment:

- Symptomatic relief only - paracetamol or ibuprofen
- Reassurance

Hepatitis A

Transmission → Faecal oral route

Presentation

- Pruritus
- Jaundice
- Prodromal symptoms like myalgia arthralgia
- Icterus

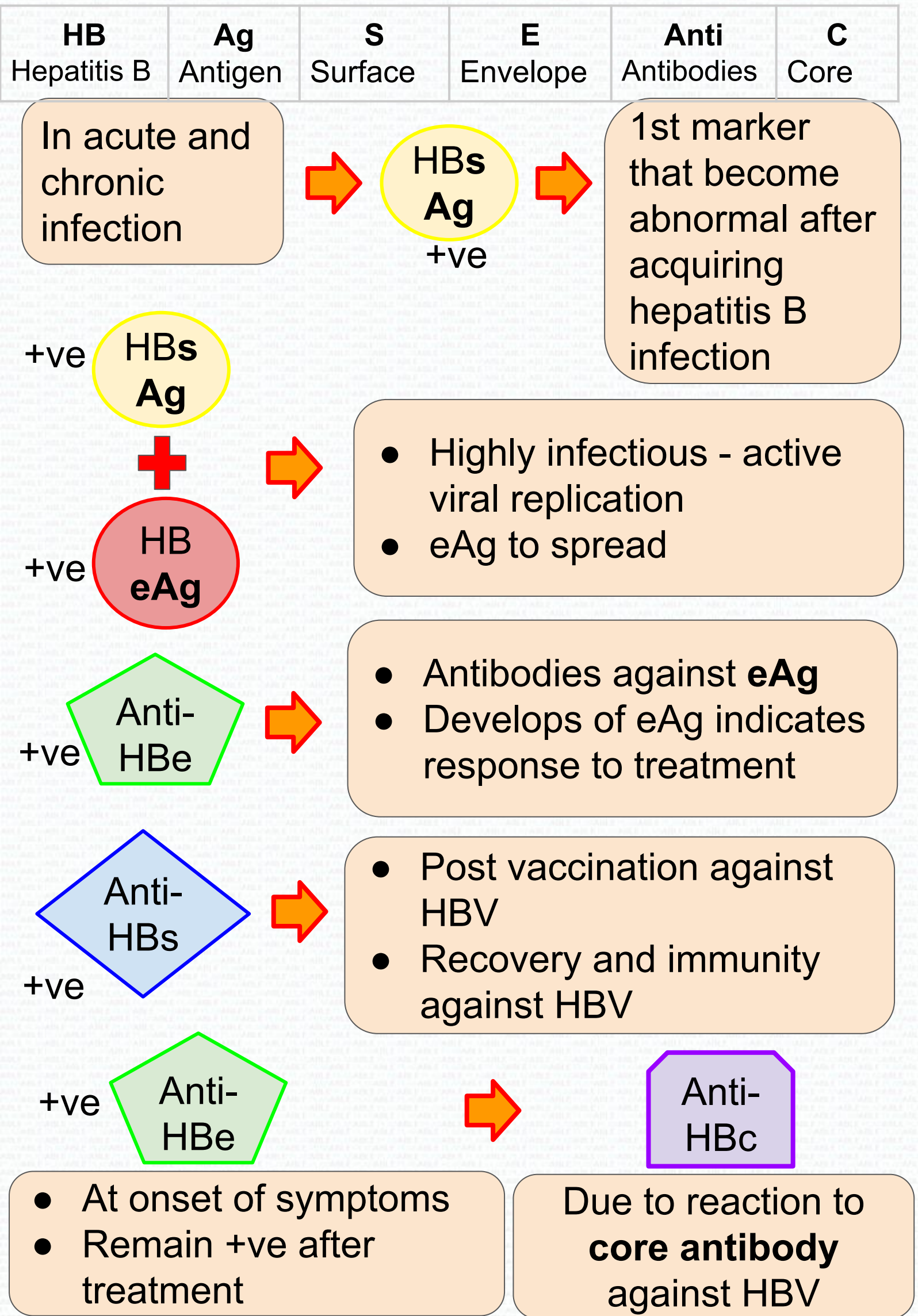
Investigations

- ALP much higher than AST
- Anti HAV **IgM** antibody → Detected around the time symptoms develop
- Anti HAV **IgG** antibody → Detectable soon after IgM and remains detectable for life

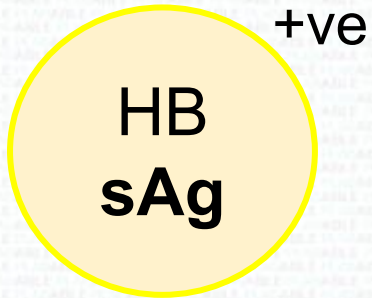
Tip

If suspect acute hepatitis A infection
→ Request IgM antibody

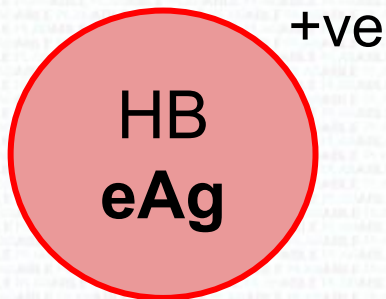
Hepatitis B Serology



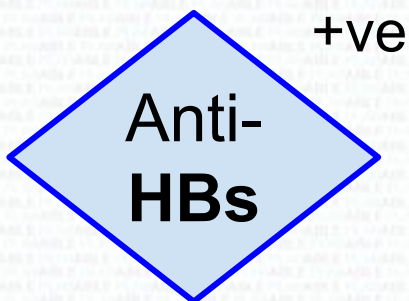
Hepatitis B Serology



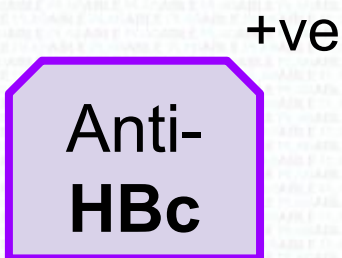
First marker to become abnormal in both acute and chronic infection



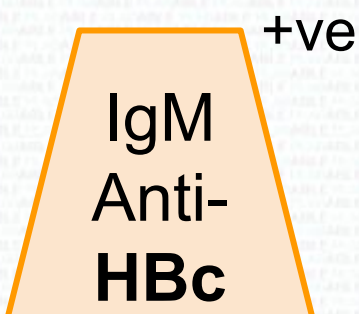
Indicates highly infectivity



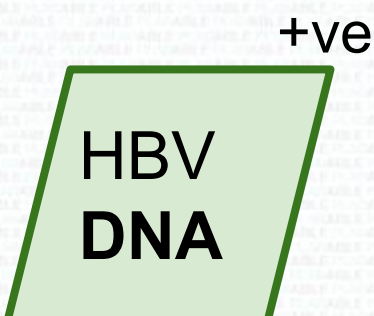
Indicates recent vaccination



Indicates past infection and remain positive after recovery

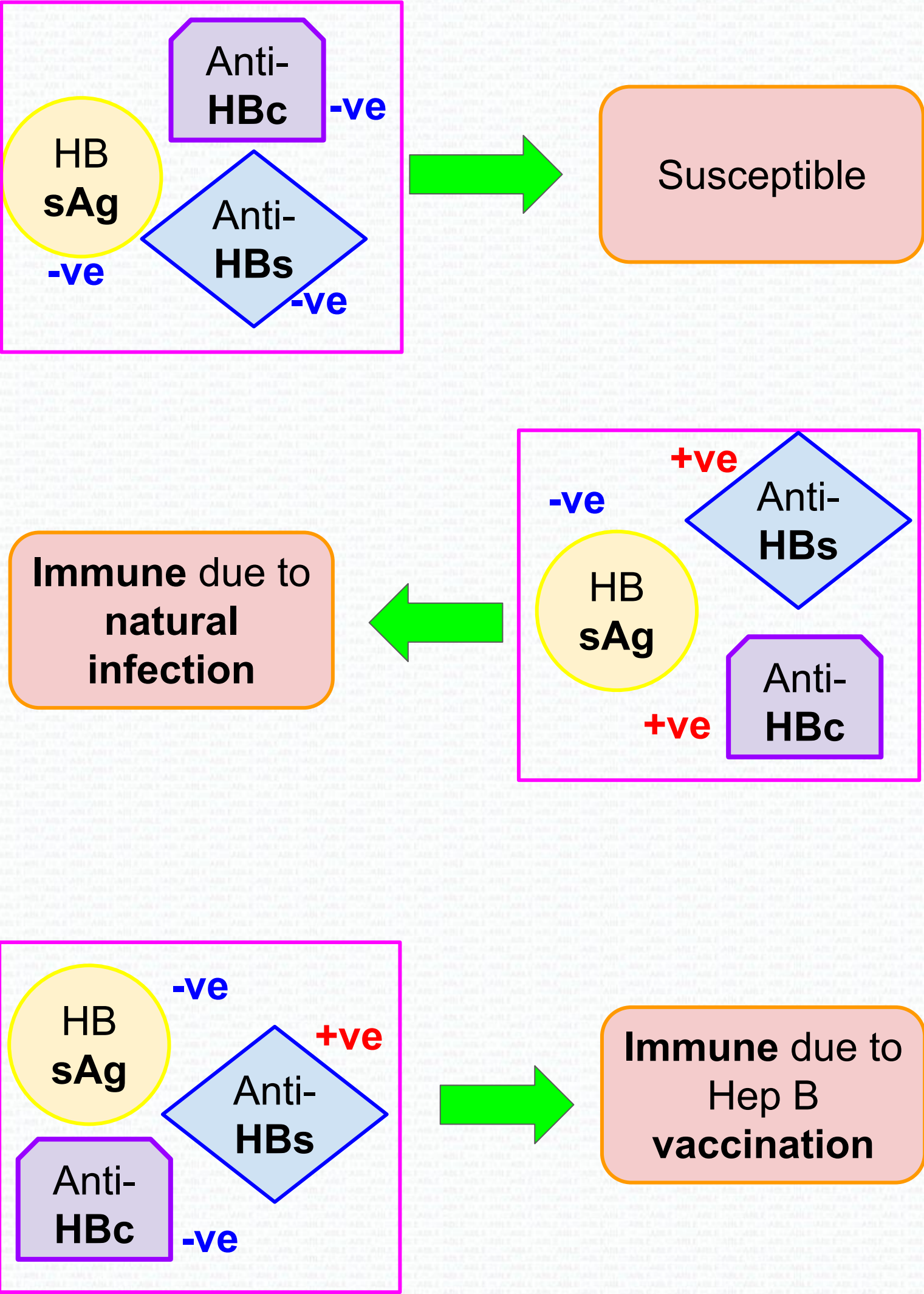


Indicates recent acute infection

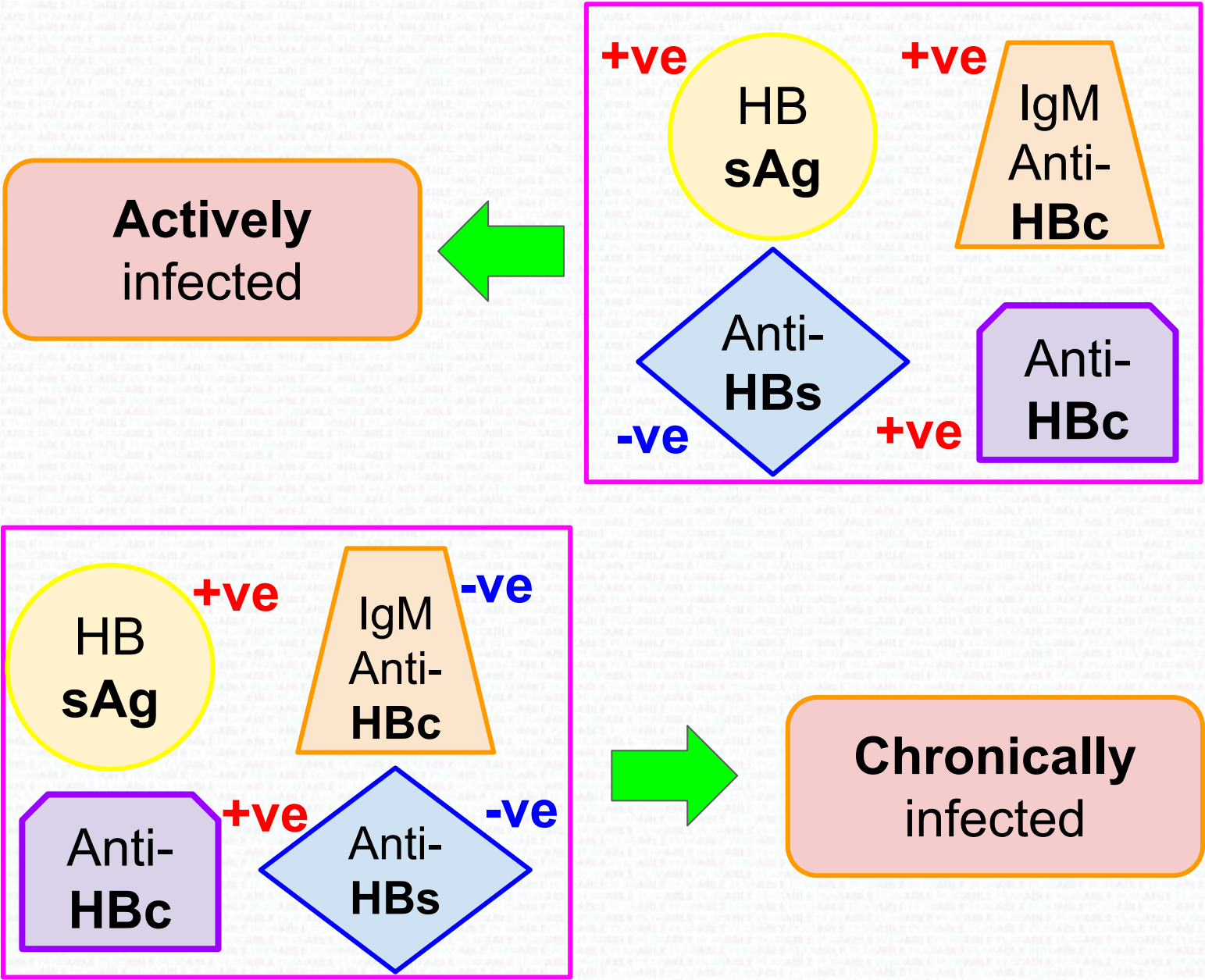


Shows infectivity (Acute viral replication)

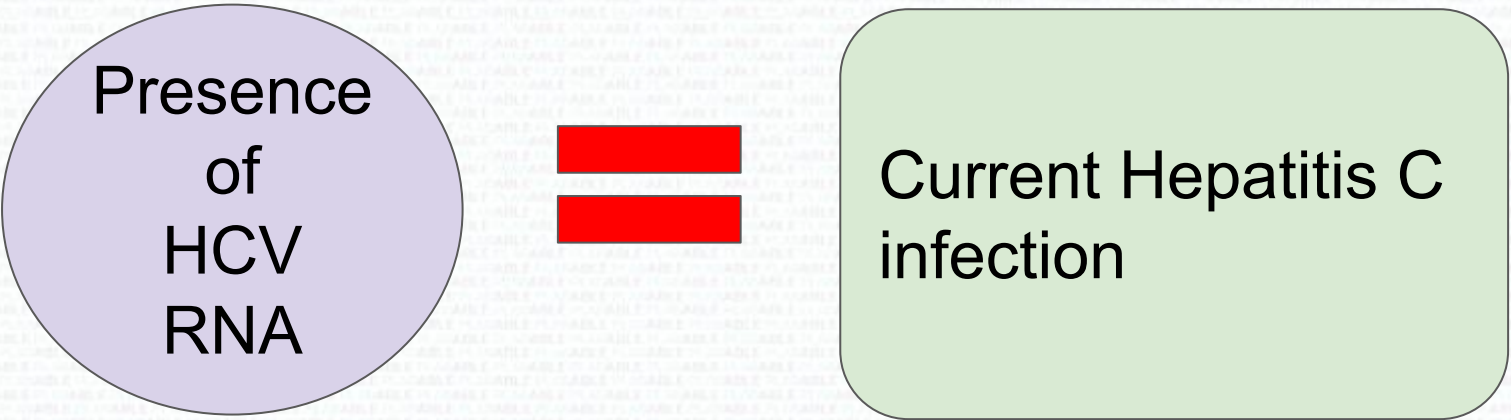
Hepatitis B Serology



Hepatitis B Serology



Hepatitis C



Hepatitis B Serology

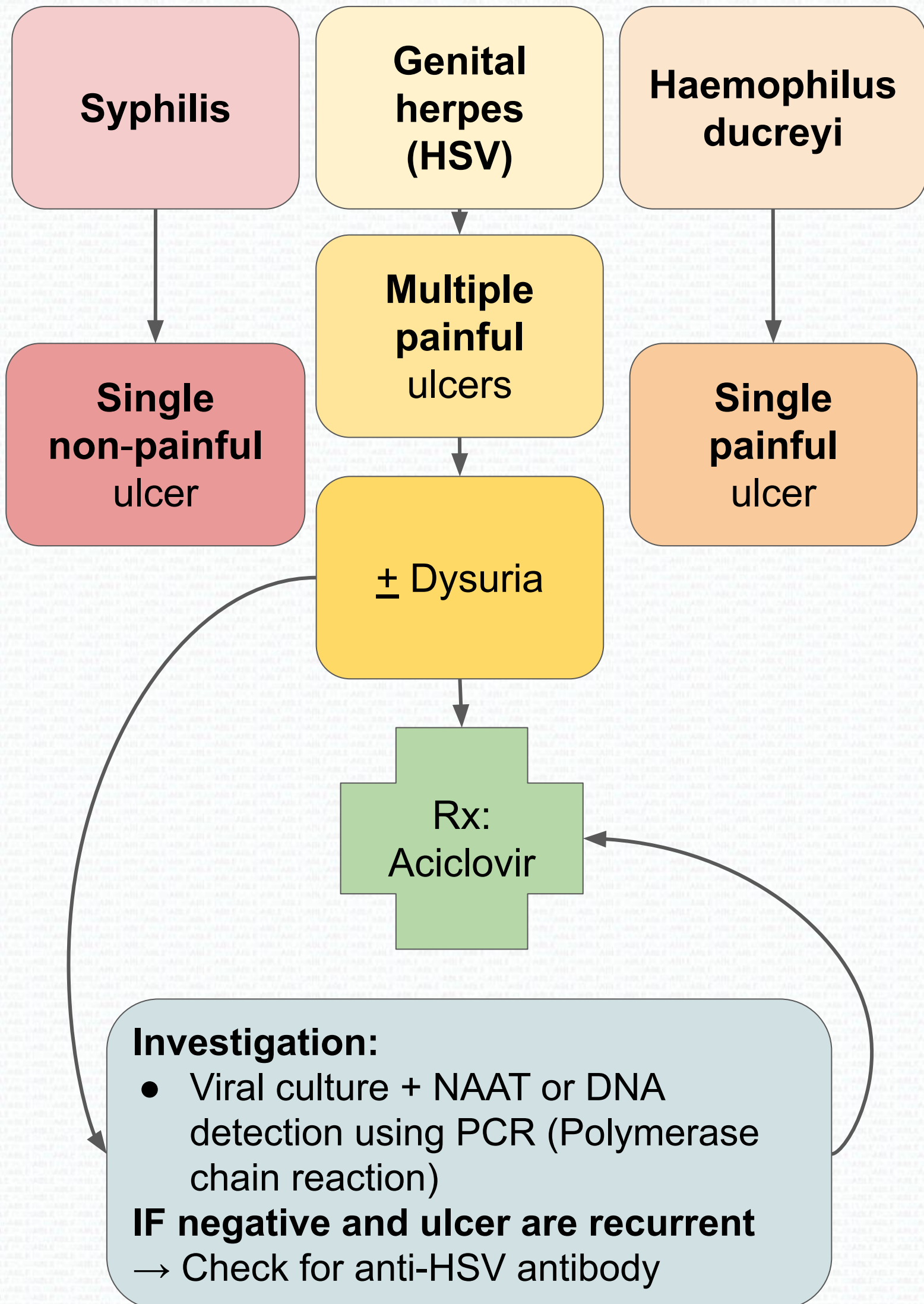
Brain trainer:

What blood result confers successful vaccination for hepatitis B?

→ **Positive: anti-HBs**

→ **Negative: HBsAg, anti-HBc**

Genital Ulcers



Type of Test

Brain trainer:

A man presents with a painful shallow ulcer on the penis. It is associated with painful inguinal lymphadenopathy. He is sexually active. What investigations is indicated?

→ Swab for haemophilus ducreyi

Type of Test

Brain trainer:

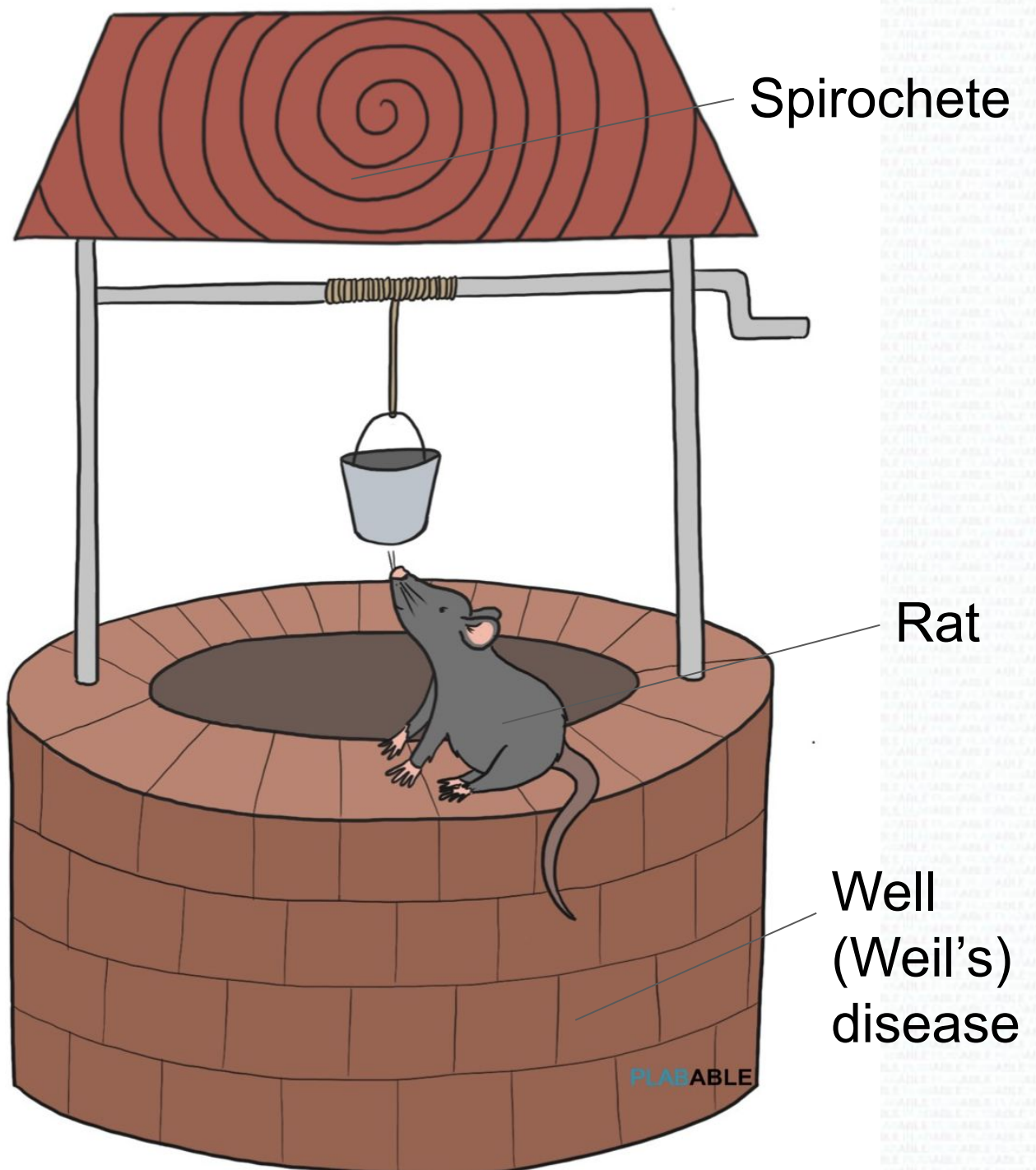
What diagnostic tests are utilised for diagnosing a primary herpes simplex virus infection?

1. Nucleic acid amplification test
2. PCR-based method
3. Viral culture

Leptospirosis

Leptospirosis

Ask yourself, what do you see in this picture that reminds you of weil's disease?



Campylobacter Jejuni

Facts:

- It is gram -ve on stool culture and sensitivity
- It is a gram -ve curved bacilli 'rods'

Features:

- Hx of travel
- Prodrome (initially): **High fever 40°C**
- Watery diarrhoea at start
- Headache
- Myalgia
- **Bloody diarrhoea after**

Treatment:

- It is mostly self-limiting with good hydration
- Abx in severe cases:
 - Erythromycin or clarithromycin or azithromycin
 - 2nd line - ciprofloxacin

Notes on other suspected infections:

- Cholera is gram -ve comma-shapes
- *Streptococcus pneumoniae* is gram +ve diplococci
- *Staphylococcus aureus* is gram +ve and coagulase +ve cocci 'round'

Travellers' Diarrhoea

Salmonella

Shigella

Campylobacter

All are gram -ve bacilli
'rod'

It is self-limiting, not treatment required
BUT for **elderly** or **immunocompromised**

Treatment for
salmonella:

- Ciprofloxacin

Treatment for
campylobacter:

1st line:

- Erythromycin
- Clarithromycin
- Azithromycin
-

2nd line:

- Ciprofloxacin

Leptospirosis

Facts:

- It is spread by contact with urine of infected animals (direct transmission)
- Or water that has been contaminated by urine of infected animals (indirect method)

Features:

- Hx of travel and water exposure
- ± Animal contact
- Presented initially with red eyes (subconjunctival haemorrhage)
- Then yellow eyes (jaundice)
- Fever, rigors, malaise, arthralgia, myalgia

Investigations:

- **Blood and urine culture and sensitivity**
- Detectable in:
 - Blood in the first **7-10 days** of disease
 - Urine after **7 days and up to 30 days**

Treatment:

- Self-limiting
- Oral doxycycline for mild cases
- Ampicillin or benzylpenicillin for severe cases

Brucellosis

Facts:

- It is most commonly spread by inhalation from infected livestock in endemic areas
- Or skin contact by veterinarians or abattoir workers

Features:

- Hx of travel to endemic area
- Animal or raw meat contact
- Presents initially with flu-like symptoms
- lymphadenopathy, splenomegaly, hepatomegaly
- arthritis, spinal tenderness

Investigations:

- Presumptive diagnosis by serum agglutination or rose Bengal test
- Direct isolation of *Brucella* spp (Gold standard)

Treatment:

- Simple infection - doxycycline
- Otherwise - doxycycline plus streptomycin

Cerebral toxoplasmosis

Facts:

- Caused by ***Toxoplasma gondii***
- Lives and reproduces in Cat's guts
- **Reactivated** in patients with **HIV infection** when **CD4 is low (<100)**

Features:

- **Increased intracranial pressure**
- Headache
- Eye pain
- Seizures
- Focal neurologic deficits
- Confusion
- Visual disturbances
- Facial Weakness

Investigations:

- Brain MRI
- → Shows ring enhancing lesion(s)

Treatment:

- Pyrimethamine
- Sulfadiazine

Infectious Mononucleosis (Glandular fever)

Facts:

- Caused by **Epstein-Barr Virus**

Features:

- Sore throat
- Fever
- Malaise
- Cervical lymphadenopathy
- Exudates of tonsils

Investigations:

- Heterophile antibodies:
 - Paul Bunnell
 - Monospot
- EBV specific antibodies

Treatment:

- Supportive measures e.g. paracetamol and ibuprofen for pain / fever

Important notes:

- Ampicillin / amoxicillin can lead to development of pruritic maculopapular rash

Summary Of Common Infections Related To Travel

India

- Fever
- Cough
- Cervical lymphadenopathy
- Caseating granuloma in LN
- **Tuberculous lymphadenitis**

India

- Flu like symptoms
- Enlarged anterior cervical LNs
- **Diphtheria**

Africa e.g. Sudan

- Intermittent fevers
- Rigors
- Malaise
- Headache
- **Malaria**

Africa

- Meningitis-like symptoms
- Anaemia
- **Cerebral malaria**

South America

- Severe headache
- Adopts a crouching position
- **Typhoid**

Far East Asia e.g. Indonesia, Thailand

- Fever
- Headache
- Retro-orbital pain
- General rash
- **Dengue fever**

Febrile Neutropenia (Neutropenic sepsis)

Facts:

- Mainly after initiation of chemotherapy
- This is due to bone marrow suppression which led to ↓ blood cells production
- Can also caused within 1 year of bone marrow transplantation

Features:

- Fever - $\geq 38.5^{\circ}\text{C}$ or 2 consecutive temperature $\geq 38^{\circ}\text{C}$
- Neutrophil count $\leq 0.5 \times 10^9/\text{L}$

Treatment:

- Start empirical antibiotic **IMMEDIATELY**
- IV tazocin (tazobactam + piperacillin)
- If patient still present with feature(s) above after **48 hours**:
 - Meropenem $\pm$ vancomycin
- If patient remain unwell after **4-6 days**:
 - Investigate for fungal infection
 - Continue antibiotics
 - Add IV antifungal

Start IV Abx in all patients who are unwell with fever and have recent chemotherapy, regardless of neutrophil count

Necrotising Fasciitis

Facts:

- Mainly caused by **group A beta-hemolytic streptococci**
- An infection spread deep and involves deep layers (dermis, subcutaneous tissues, fascia and muscles)
- Life threatening and rapidly spread into deep layers
- IM and SC injections, immunosuppression and diabetic are risk factors

Features:

- Presented as cellulitis initially (first 1-2 days)
 - Erythema
 - Swelling
 - Pain over the affected area(s)
- Then it will be presented as bullae
 - Grey/black skin (necrosis)
 - Hard subcutaneous tissue
- Eventually septic shock
- Severe Pain

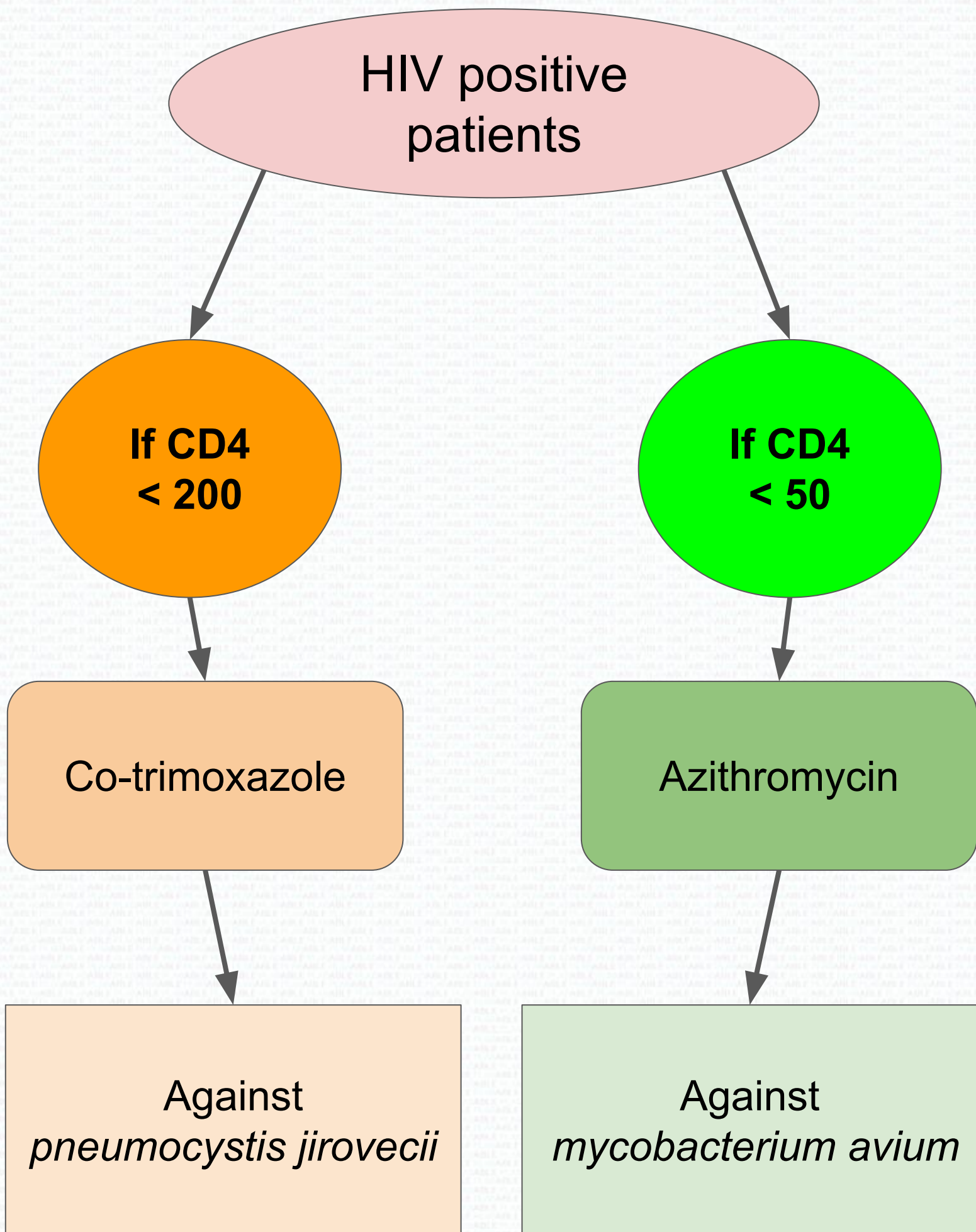
Treatment:

- Urgent surgical debridement
- IV antibiotics - clindamycin or benzylpenicillin
 - *This condition **DOES NOT** respond to flucloxacillin*

Differ from erysipelas as necrotising fasciitis is **diffuse and deep**.

Erysipelas is **well demarcated** infection

Prophylaxis Antibiotics For HIV Positive Patients



Abscess

Features:

- High fever
- Erythematous skin swelling
- Neck abscess if large enough can cause dysphagia

Treatment:

- IV antibiotics
- Incision
- Drainage

Key point to **THINK SEPSIS**

- Unresponsive or response to only voice/pain
- Acute confusion state
- **Systolic blood pressure ≤ 90 mmHg**
- **Heart rate ≥ 130 /min**
- **Respiratory rate ≥ 25 /min**
- Require oxygen support to keep $\text{SaO}_2 \geq 92\%$
- Non-blanching rash
- Mottled, ashen, cyanotic
- Not passed urine over last 18 hrs or **urinary output < 0.5 mL/kg/hr**
- **Lactate ≥ 2 mmol/L**

IV antibiotic ASAP if sepsis, NOT oral

Whipple's Disease

Facts:

- It is a rare **multisystem disorder**
- Caused by **tropheryma whippelii** infection

Features:

- **Malabsorption** - weight loss and diarrhoea
- Large joint arthralgia
- Lymphadenopathy
- Skin hyperpigmentation and photosensitivity
- Pleurisy
- Pericarditis
- Neurological symptoms (rare):
 - Ophthalmoplegia
 - Dementia
 - Seizure
 - Ataxia
 - Myoclonus

Investigations:

- Jejunal biopsy shows:
 - Stunted Villi
 - Deposition of **macrophages containing Periodic acid-Schiff (PAS) granules**

Treatment:

- Co-trimoxazole for a year
- $\pm$ preceded by course of IV penicillin

Important Investigation Result For Diagnosis

Whipple's Disease:

- Jejunal biopsy shows:
 - Deposition of macrophages containing periodic acid-Schiff (PAS) granules

Celiac Disease:

- Jejunal/duodenal biopsy shows:
 - Villous atrophy 'shortening'
 - Crypt hyperplasia
 - Lymphocytosis



Lymphoma:

- Patient with **known celiac disease**
- Duodenal biopsy shows:
 - Lymphomatous infiltrates

T-cell lymphoma is a rare complication of celiac disease

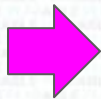
Mastitis / Breast Abscess

- Commonly caused by Staph. Aureus
- Via breast feeding

Notes on breastfeeding

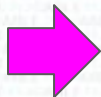
If mother has:

HIV



AVOID breastfeeding

Breast abscess



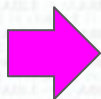
Continue breastfeeding

Nipple Candidiasis



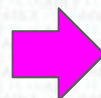
Continue breastfeeding

Hepatitis B



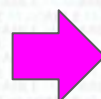
Continue breastfeeding
(if baby has received Hep B vaccine)

Hepatitis C



Continue breastfeeding
(unless nipples is cracked or bleeding)

Tuberculosis



Continue breastfeeding
(Baby needs BCG vaccinated ASAP)

Depression



Continue breastfeeding
(if mother is on Sertraline)

Rabies

Brain trainer:

What is the only circumstance in which rabies vaccination is indicated in the United Kingdom?

→ **Bat bite**

Neither domestic nor wild animals (aside from bats) are carriers of the rabies virus in the UK

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